Answer keys and instructor notes

Critical appraisal workshop

Instructor only

Pre and post quiz

15 questions, same sheet at the start and at the end (pack p. 2-3 and 26-27).

Quiz answers

Q	Question	Ans	Layer	Note
1	RR 0.95 (0.80 to 1.13): best conclusion	B	7	No difference shown; 20% reduction to 13% increase compatible. Absence of evidence is not evidence of absence
2	Registry 90-day mortality, paper 28-day primary	B	3, 5	Outcome switching: disclose and justify
3	Main purpose of a SAP finalised before unblinding	C	3	Limits analytical flexibility after the data are seen
4	12 untreated patients excluded, called ITT	B	6	ITT analyses patients as randomised
5	OR 1.40 (1.05 to 1.87), p = 0.21	C	4	CI excluding 1 implies $p < 0.05$; true p about 0.02
6	n = 17, mean of an integer = 2.30	B	4	GRIM: 39/17 = 2.29, 40/17 = 2.35
7	Best design for a causal drug claim	B	1	Well-conducted RCT
8	Guideline for an ML prediction model	C	1	TRIPOD+AI; CONSORT-AI is for trials of AI interventions
9	RoB tool, non-randomised comparison	B	6	ROBINS-I
10	Null primary, abstract highlights a subgroup	B	7	Spin
11	Paper-mill signal	B	5	Tortured phrases (Cabanac 2021)
12	Confidential manuscript and AI tools	B	AI	Never upload; follow journal policy; disclose permitted use
13	LLM says the registry lists another primary	C	AI	Open the registry yourself, incl. history of changes
14	Unblinded assessment biases most	B	6	Subjective outcome rated by the treating team
15	A major issue should contain	B	Verdict	Problem, why it matters, concrete request

Report the mean score before and after (paired) and the per-question change at the wrap-up or in the follow-up email. Note: 12 of 15 answers are B.

Exercise 1 Open science

TTM2 and Gautret traces, scorecard. COMPare is now covered on slide 9 only.

Part A. TTM2 trace

Registration	16 September 2016; first enrolment 18 November 2017: **prospective**
Primary outcome	all-cause death at 6 months, identical in registry, protocol, SAP and paper
Main secondary	poor functional outcome (mRS 4 to 6) at 6 months
Sample size	1200 raised to 1900 for TAME co-enrolment, before any results: transparent
Effect size	SAP: 55% control mortality, 7.5 points absolute, 13.6% relative; paper says "15%" (7.5/50)
SAP	published in *Trials* October 2020, after enrolment ended (January 2020), team still blinded: check finalisation date vs unblinding
Verdict layers 1-3	all OK; the only note is the RRR wording

Part B. Gautret: EU register 2020-000890-25 vs paper

Item	Register	Paper
Controlled?	**No**	compares with 16 controls (other centres, refusers)
Randomised?	No	No
Planned n	25 (France)	42 enrolled, 36 analysed
Primary end point	virus detection at **days 1, 4, 7 and 14**	PCR at **day 6**
Analysis	not stated	interim, shared early "because our first results are so significant and evident"

Q4 unplanned interim analysis, no stopping rule. **Q5** protocol or SAP should define handling of withdrawals and ICU transfers. **Q6** layer 5 context (independence of review): received 16 and accepted 17 March 2020; an author was editor-in-chief. Does not prove the result wrong; removes independent scrutiny. Retracted December 2024.

Register fields E.5.1, E.5.1.1, E.8.1, F.4.1, checked 11 October 2026.

Part C. The scorecard

Q7 data sharing (and actually obtainable data). Reviewers see summary tables only.

Item	TTM2	Gautret
Registered before first enrolment	Yes	Cannot tell (approved 5-6 March 2020)
Protocol public before results	Yes (2019)	No
SAP public before unblinding	Yes (2020)	No
Primary outcome unchanged	Yes	No (day 6 vs days 1/4/7/14)
All registered outcomes reported	Yes	No
Data-sharing statement	Yes (on request)	No
Individual patient data (IPD) obtainable	Partly (on request)	No
Code available	No	No
Independent peer review	Yes	No

Exercise 2 Methodology stations

Estimand, CI and p, forensics, bias and spin.

Station 1. Estimand

	TTM2	Gautret
Population	comatose adults after OHCA, presumed cardiac or unknown cause	hospitalised patients, one centre (controls elsewhere)
Treatments	33 °C vs normothermia with early fever treatment	HCQ (± AZ) vs no HCQ, **not assigned**
Variable	death by 6 months	PCR negative at day 6 (surrogate; register said 1/4/7/14)
Intercurrent events	treatment policy	ICU, death, discharge, stop: **excluded** (outcome-dependent)
Summary	relative risk	proportions per day, many p-values

Q3 answers "are treated patients in Marseille who stayed on ward different from untreated patients elsewhere?", not "what does HCQ do". **Q4** 7.5/55 = 13.6%; the paper's 15% uses 50%. Minor, but a reviewer notes it.

Stations 2 and 3. Numbers

	TTM2	Gautret
S2 Q1 percentages	465/925 = 50.3% vs 446/925 = 48.2%	
S2 Q2 p from the CI	SE = (ln 1.14 − ln 0.94)/3.92 = 0.049; z = 0.80; p ≈ 0.42: compatible with 0.37 (adjusted model, rounded limits)	
S2 Q3 one sentence	Correct: "no difference shown; from a 6% reduction to a 14% increase compatible". Wrong: "hypothermia does not work"	
S2 Q4 range	RR 0.94 to 1.14: a 6% reduction is **not** ruled out	
S2 Q5 multiplicity		4 days × several comparisons: a dozen or more tests, no correction, n = 36; any single p, even 0.001, loses its meaning
S3 A/B flow	1900 − 37 no consent − 2 randomised twice = **1861** ITT (930/931); − 11 missing primary (5/6) = **1850**; other denominators must be explained	42 → 26 HCQ + 16 controls; 6 HCQ removed (3 ICU, 1 death, 1 discharged, 1 nausea) → **20 vs 16**; all losses in the treated arm, all bad outcomes: bias towards benefit

S3 C (fictional). **1** 2.41 × 13 = 31.33: impossible (GRIM). **2** CI excludes 1, p ≈ 0.04: 0.11 inconsistent. **3** 87/120 = 72.5%. **4** 22 p-values 0.62 to 0.98: too similar (Carlisle); ask.

Station 4. Bias and spin

ROBINS-I (Gautret)	
1 Confounding	**Critical**
2 Selection	Serious-Critical
3 Classification	Low-Moderate
4 Deviations	Serious
5 Missing data	**Critical**
6 Measurement	Moderate-Serious
7 Reported result	Serious
Overall	**Critical**

RoB 2 (TTM2): Low in all five domains. Unblinded clinicians matter more for functional outcome and for withdrawal of life support (a path to death); mitigated by blinded, protocolised prognostication.

Spin in Gautret conclusion: "significantly associated" framed as effect; "reinforced by azithromycin" from 6 patients; "despite its small sample size" then strong claim.

Honest rewrite: "In this non-randomised study of 36 patients, with 6 treated patients excluded after ICU transfer, death or withdrawal, day-6 PCR negativity was more frequent in treated patients. The design cannot separate treatment from centre and selection effects; randomised trials are needed."

Exercises 3 to 5 Review and LLM

Planted errors in manuscript A, pair report, audit rules, generic vs structured prompt experiment, notes.

Manuscript A: the eight planted errors

#	Category	Error
1	External: outcome switch	FAC 4+ at discharge reported as primary; registered day 30 ($p = 0.21$)
2	Design: false ITT	8 early-group patients excluded, still called ITT
3	Internal: n randomised	abstract 324; results 320
4	Internal: percentage	female 118/160 = 73.8%, printed 71.5%
5	Internal: p vs CI	delirium RR 0.70 (0.47 to 1.04), $p = 0.02$; true $p \approx 0.07$
6	Internal: GRIM	$n = 19$, mean 3.48: impossible
7	Design: unblinded assessment	FAC rated by the treating physiotherapist; registry promised blinded
8	Interpretation: spin	"restores independent walking ... standard care for all"

Not errors: 152 analysed vs 153 required per arm; falls and death RRs and CIs are correct; the primary RR and CI as printed are correct (the problem is *which* outcome is primary). Valid extra criticisms (multiplicity, MI not shown) count as neither.

Manuscript B (instructor experiment only): answer-keys/llm-exercise-key.md.

The 7-layer grid applied: three verdicts

	TTM2	Gautret	Manuscript A
Start	High (RCT)	Moderate (non-randomised)	High (RCT)
1-2 Type, design	OK (RRR 15% vs 13.6%: note)	**Serious**: causal claim, external controls, surrogate	OK (152 vs 153 is not an error)
3 Pre-specification	OK	Concern: no public protocol or SAP	OK: registered, SAP signed before enrolment
4 Internal	OK	OK: numbers add up (exclusions are layer 6)	**Serious**: E3-E6 (n, %, p vs CI, GRIM)
5 External	OK	**Serious**: day 6 vs days 1/4/7/14; uncontrolled in register	**Serious**: outcome switch (E1), blinding promise
6 Validity	OK (RoB 2 low)	**Serious**: ROBINS-I critical	**Serious**: false ITT (E2), unblinded assessor (E7)
7 Interpretation	OK	**Serious**: spin	**Serious**: spin (E8)
Confidence	**High**	**Very low** (moderate, down two levels); claim not supported as stated	**Very low** (layer 4); low from layers 5-6 alone
Recommendation	Accept or minor revision	Reject: design cannot answer the causal question	Major revision at best: registered primary (null), true ITT, corrected numbers, new conclusion

Teaching point: both very low verdicts, different layers, different recommendations. Gautret fails on design; A fails on conduct and reporting, partly fixable by revision. Gautret's editorial independence goes to confidential comments, not into confidence.

Exercise 4. What a good pair report looks like

Summary (3 lines): RCT, early vs standard mobilisation after hip fracture, n = 320; claims better walking independence at discharge and less delirium.

Major issues, ranked by impact, each as problem / why it matters / request:

1. **Outcome switch (E1).** Paper primary = FAC 4+ at discharge; registry and SAP primary = FAC 4+ at day 30 (53.9% vs 46.8%, RR 1.15, 0.92 to 1.44, p = 0.21), undisclosed. The positive headline rests on a switched outcome. *Request:* report the registered primary as primary, state and date the change, rewrite abstract and conclusion.
2. **Not ITT (E2).** 8 early-arm patients excluded after randomisation, one arm only, likely the frailest. *Request:* true ITT on all 320, exclusions as sensitivity analysis.
3. **Unblinded assessor (E7).** FAC rated by the treating physiotherapist; registry promised blinding. *Request:* explain the deviation, discuss detection bias.
4. **Numbers do not add up (E3 to E6)**: 324 vs 320, 71.5% for 118/160, delirium p vs CI, mean 3.48 impossible for n = 19. *Request:* recheck all data; drop the delirium claim.
5. **Spin (E8)**: "restores", "standard care for all". *Request:* conclusion matching the registered primary.

Minor: MI results not shown, deaths in FAC not stated, no adjustment for stratification factors.
Recommendation: major revision at best (see the three verdicts).

Exercise 5 Part B. Scoring the LLM output (audit rules)

Accept as a detection (count as found):

- E1: any mention that the primary outcome or time point differs from the registry or SAP, or that the null day-30 result was the registered primary. A boilerplate "check the registry" does not count.
- E2: exclusions incompatible with ITT, or post-randomisation exclusion biases the result. E3: abstract count differs from results or registry.
- E4: the specific wrong percentage (71.5%, correct 73.8%). E5: delirium p inconsistent with the CI (true p = 0.07).
- E6: mean 3.48 impossible, cannot come from integers (66/19 = 3.47, 67/19 = 3.53).
- E7: assessor not blinded and/or deviates from the registry. E8: conclusion overstates or is not supported by the registered primary.

Valid extras ("Added" if new, never "Wrong"): no multiplicity correction for secondaries (allowed by the SAP); MI results not shown; sample size only refers to the SAP; length of stay p without a named test; no adjustment for stratification factors; single-country generalisability; death counted as not walking not stated; discharge as a treatment-dependent time point.

Not errors (mark "Wrong" if the LLM claims them): 152 analysed vs 153 required per arm (adequate within rounding); falls RR and CI correct; death RR and CI correct; primary RR and CI as reported are correct (the problem is which outcome is primary). Any invented registry detail, reference or recomputed number that does not check out is "Wrong".

Pilot: LLM alone on manuscript A

	LLM 1	LLM 2	LLM 3
Model and version, date	Claude Opus 5.5, 2026-10-11		
Planted errors found (mean of 5 runs) / 8	8.0 (8 in all 5 runs)		
Most often found	all 8, every run		
Most often missed	none		
Wrong claims (invented registry, refs, arithmetic)	0; no trap flagged		

Claude: 5 fresh sessions, no tools, fixed prompt, raw outputs and scoring in instructor/llm-experiment/pilot/. Recomputed delirium p as 0.07 to 0.08; valid extras every run (discharge as treatment-dependent time point, MI not shown, deaths in day-30 FAC). Pilot prompt was semi-structured and included the sources, hence the ceiling; the 2x2 experiment on the next pages separates prompt from sources. Participants report their own LLM results live in the reveal (columns 2 and 3).

Experiment: generic vs structured prompt, with and without sources

Arm (5 runs each)	A: found /8	B: found /8	Missed
G, manuscript only	6.6 (E1? 1/5)	7.0 (E1? 2/5)	E1 (needs registry); E6 GRIM in 2 A runs
G, + registry + SAP	7.8	8.0	E6 GRIM in 1 A run
S, manuscript only	7.0 (E1? 5/5)	7.0 (E1? 5/5)	E1 (needs registry)
S, + registry + SAP	8.0	8.0	none
Wrong claims	0	0	no trap flagged in 40 runs
Words per review	G 1640, S 2680	G 1570, S 2710	S = 60 to 80% more to audit

Claude Opus 5.5, 2026-10-11, headless, no tools, no memory, fresh session per run; inputs stripped of any hint that the paper is fictional. Prompt G and S as in worksheet 03. E1? = suspected: the run named the null outcome (A day 30 FAC, B any SSI at 30 days) as the possible registered primary; a boilerplate "check the registry" does not count. Raw runs, inputs and scoring: instructor/llm-experiment/.

Experiment: messages for the reveal

1. **Documents beat prompts.** The outcome switch was confirmed in 20/20 runs with the registry and SAP, 0/20 without. No prompt can find what is not in the input.
2. **Structure decides the suspicion.** Without sources, the structured prompt raised the right suspicion ("was day 30 the registered primary?") in 10/10 runs, the generic prompt in 3/10. Every S run listed the checks it could not do.
3. **Arithmetic is near ceiling either way.** The only misses (GRIM, 3/20 A runs) were all with the generic prompt: the model saw the odd sentence but did not test it.
4. **Where did the structure come from?** From the method of part 1 and the reporting guideline. The prompt is only as good as the reviewer who writes it.
5. **Zero hallucinations here is not a guarantee.** Clean planted errors, short paper, a strong model. Real manuscripts have ambiguous problems; free tools and other models may do worse. Verify every point.
6. **Cost.** S reviews are longer: more to audit. And real manuscripts under review must never be pasted into an LLM.

Limitation: the manuscript-only input starts at "Document 3", a hint that documents 1 and 2 exist.

Exercise 5 Part C. Suggested answers to the discussion

1. **Layers covered and skipped.** Expect layers 4 and 7 well covered (numbers, spin are visible in the text). Most skipped: layers 3 and 5 (needs the registry and SAP: E1, E7 promise) and layer 6 subtleties (who assessed the outcome, E7). Link back to the grid: a layer you do not open, you do not find.
2. **Arithmetic: you or the LLM?** Usually the LLM: near ceiling in our runs (E3 to E6 found almost every time; only misses were GRIM, 3/20 A runs, generic prompt). Humans under time pressure skip recomputation. But every recomputed number must still be checked: you audit, it does not.
3. **Needs a second document.** Without the registry and SAP, nobody can confirm E1 (0/20 LLM runs) or the broken blinding promise; at best a reviewer suspects it (S prompt 10/10, G 3/10). Lesson: fetch the registry and SAP first; no prompt finds what is not in the input.
4. **Prompt G vs S.** S raised the right suspicion, listed the checks it could not do, and missed no arithmetic; with sources both near 8/8 (G 7.8, S 8.0). Cost: S reviews 60 to 80% longer to audit. The structure came from part 1: the 7 layers and the reporting guideline. The prompt is only as good as the reviewer who writes it.

Close with: 0 wrong claims in 40 runs is not a guarantee; never paste a real confidential manuscript.

Notes for the "When AI conducts research" slide

PNAS 2026 (Brodeur et al.). 288 researchers, 103 teams, randomised: human-only 94%, AI-assisted 91%, AI-led 37% reproduction. Human-only found significantly more major coding errors. AI-assisted gave no measurable advantage. Caveat: social-science computational reproduction; the authors call AI-led a lower bound of AI capability.

Backup, no longer on the slide: **REPRO-Bench (Hu et al., Findings ACL 2025).** 112 social-science papers with public reproduction reports; best of three agents 21.4% accuracy.

Luo, Kasirzadeh, Shah (arXiv 2025, preprint). Four failure modes in AI scientist systems: benchmark selection, data leakage, metric misuse, post-hoc selection. Logs and code reveal them far better than the paper: an open-science argument (link back to block 1).

Lu et al. (Nature 2026). The AI Scientist's paper passed first-round review at an ML workshop (70% acceptance). Use it as a peer-review point: reviewers did not detect it; the authors themselves flag overloaded review systems and noise in the literature. Do not claim more than the abstract without reading the full text.

Wu et al. (JMIR 2026). Claude agent (Chat, Code, Cowork modes), public Moorfields AMD dataset, 7802 eyes, 27 runs. 9/9 SAPs right framework; KM estimates near-identical; systematic execution errors repeated across runs; SAP quality did not predict code correctness; one undisclosed rerun after a crash; 8/17 summaries fully satisfactory, 2 clinically wrong. Note to the room: the agent is from the same family as some tools participants may use.

Three levels box. If time allows, ask: an agent reports Cox HR 0.72, $p < 0.05$, no errors; what would you check? Expect time zero and confounding first, PH and the estimand question last.